Extra Credit: On the Compaction Schnorr Signature Scheme

Let (S1, R1), ...,(Sn, Rn) be n distinct Schnorr signatures computed on distinct messages M1, M2, ..., Mn under the same private key Ks. Is it possible to aggregate these signatures and then broadcast it as one single-compact (constant-size) signature to be verified by all the signers?

If your answer is YES, show mathematically (and algorithmically) how it can do. If your answer is NO, similarly, how mathematically (and algorithmically) why not. Your answer should not be handwaving verbal descriptions, and it must include necessary mathematical rigor to showcase possibility or impossibility result. Note that in the class, we showed in detail how Schnorr signature is computed with a correctness verification. You can recall these equations to deduce your conclusions.

Answer:

- **No, it is not possible** to aggregate n distinct Schnorr signatures on distinct messages into a single, constant-size signature under the same private key while keeping it independently verifiable without extra data.
- **Why it fails mathematically:** From class, a standard Schnorr signature on message m uses a random per-signature nonce r to compute the auxiliary value $R = \alpha^r \mod p$ and the signature scalar $s = r - H(m \parallel R) \cdot y \mod q$.
- **The binding problem:** If we try to add up all n signature scalars into a single aggregate value like $S_{\text{agg}} = \sum s_i \mod q$, the equation expands to $S_{\text{agg}} = \sum r_i - y \sum H(M_i \parallel R_i) \mod q$.
- **Why constant size breaks down:** To verify this with a single constant-size check, the verifier needs the sum of the hashes $\sum H(M_i \parallel R_i)$. But to compute those hashes, the verifier *must* have every individual R_i value for every message M_i.
- **The trade-off:** If you drop the individual R_i values to keep the signature constant-size, the verifier cannot evaluate the hashes and verification fails. If you include all R_i values so the verifier has them, the signature size grows linearly ($O(n)$) and is no longer constant size. Therefore, true constant-size aggregation across distinct messages is mathematically impossible for Schnorr signatures.